Machine Learning Basics

Instructors: Babak n. Arabi, Mohammadreza A. Dehaqani, Mostafa Tavassolipour

Hamed Soltani, Parisa Mohammadi

Fall 2025

Solution of Homework 3

Question 1: Answer These Questions

1. Why is the dual formulation of SVM usually preferred when kernels are used? Provide two reasons.
2. Explain the statement: "Only support vectors influence the final SVM classifier." Why do non-support vectors have zero dual coefficients?
3. Explain mathematically why the kernel trick allows SVMs to operate in infinite-dimensional spaces without computing the feature map explicitly.
4. Explain how the choice of kernel (linear, polynomial, RBF) affects the decision boundary, model capacity, and tendency to overfit.

Answer

(a)

(1) In the dual formulation, data appear only through inner products $\langle x_i, x_j \rangle$, which can be replaced by a kernel function $K(x_i, x_j)$ without explicitly mapping to a high-dimensional feature space. (2) The dual optimization depends only on the number of samples, not on the dimensionality of the feature space. Thus, even infinite-dimensional kernels (e.g., RBF) remain computationally feasible.

(b)

In the dual SVM solution,

$$w = \sum_{i=1}^{n} \alpha_i y_i x_i,$$

only samples with $\alpha_i > 0$ contribute to the weight vector. KKT complementary slackness implies that points with functional margin strictly greater than 1 must satisfy $\alpha_i = 0$. These points lie far from the decision boundary and therefore do not influence its shape. Only points on or inside the margin (support vectors) can have non-zero α_i, shaping the classifier.

(c)

Let $\phi(x)$ be an implicit feature map, possibly infinite-dimensional. The dual SVM requires only inner products:

$$K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle.$$

If $K(\cdot,\cdot)$ is positive semidefinite, Mercer's theorem guarantees that such a feature map exists, even if infinite-dimensional. Because the optimization never requires the explicit computation of $\phi(x)$, we avoid exponential or infinite computational cost while still exploiting richer hypothesis spaces.

(d)

A linear kernel yields a linear decision boundary and low model capacity, with low overfitting risk. A polynomial kernel introduces nonlinear decision boundaries whose complexity increases with the degree, raising the risk of overfitting. The RBF kernel corresponds to an infinite-dimensional feature space and can fit highly complex boundaries; it may overfit unless parameters γ and C are regularized properly.

Question 2: RBF kernel feature space intuition

Explain why the Gaussian (RBF) kernel $K(x, z) = \exp(-\|x-z\|^2/(2\sigma^2))$ can be interpreted as an inner product in an infinite-dimensional Hilbert space.

Answer

Using Taylor expansion of exponential:

$$\exp\left(\frac{x^\top z}{\sigma^2}\right) = \sum_{k=0}^{\infty} \frac{1}{k!}\left(\frac{x^\top z}{\sigma^2}\right)^k,$$

and note the Gaussian kernel can be written (after completing the square) as

$$\exp\left(-\frac{\|x-z\|^2}{2\sigma^2}\right) = \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right)\exp\left(-\frac{\|z\|^2}{2\sigma^2}\right)\exp\left(\frac{x^\top z}{\sigma^2}\right).$$

The last exponential expands into an infinite series of homogeneous polynomial terms of all degrees, which corresponds to an inner product in an infinite-dimensional feature space with appropriately weighted monomials. Thus RBF corresponds to an inner product in an RKHS of infinite dimension.

Question 3: SVM dual — symmetric 4-point example

Consider the two-class data

$$x_1 = (1,1)^\top,\ y_1 = +1, \qquad x_2 = (1,-1)^\top,\ y_2 = +1,$$

$$x_3 = (-1,1)^\top,\ y_3 = -1, \qquad x_4 = (-1,-1)^\top,\ y_4 = -1.$$

Using a hard-margin SVM with linear kernel (i.e. feature map is identity), solve the dual problem to find the optimal dual variables α_i. Then compute the primal parameters w and b.

Answer

The hard-margin SVM dual is

$$\max_{\alpha\in\mathbb{R}^4} \quad L(\alpha) = \sum_{i=1}^{4}\alpha_i - \tfrac{1}{2}\sum_{i=1}^{4}\sum_{j=1}^{4}\alpha_i\alpha_j y_i y_j\, x_i^\top x_j,$$

$$\text{s.t.} \quad \alpha_i \ge 0, \quad i = 1,\dots,4, \qquad \sum_{i=1}^{4}\alpha_i y_i = 0.$$

Compute the Gram (dot-products) for these points:

$$x_1^\top x_1 = 2,\ x_2^\top x_2 = 2,\ x_3^\top x_3 = 2,\ x_4^\top x_4 = 2,$$

$$x_1^\top x_2 = 0,\ x_1^\top x_3 = 0,\ x_1^\top x_4 = -2, \qquad \text{(and similarly by symmetry).}$$

By symmetry of the problem (two identical positives and two identical negatives mirrored across $x_1 = 0$), try the ansatz

$$\alpha_1 = \alpha_2 = \alpha_+ \qquad \text{and} \qquad \alpha_3 = \alpha_4 = \alpha_-.$$

The equality constraint $\sum_i \alpha_i y_i = 0$ becomes

$$\alpha_+ + \alpha_+ - \alpha_- - \alpha_- = 0 \quad \Rightarrow \quad 2\alpha_+ - 2\alpha_- = 0$$

so $\alpha_+ = \alpha_- =: \alpha$. Thus all four dual variables are equal: $\alpha_i = \alpha \geq 0$.

Plug this into the dual objective. Sum of α_i is 4α. For the quadratic term observe many cross-terms vanish due to zero dot-products; a direct computation yields

$$L(\alpha) = 4\alpha - \tfrac{1}{2} \cdot 16\alpha^2 \cdot \tfrac{1}{4} = 4\alpha - 8\alpha^2.$$

(Equivalently, one can compute the double sum explicitly using the dot-products and labels; the result simplifies to $L(\alpha) = 4\alpha - 8\alpha^2$.)

Maximize $L(\alpha)$ over $\alpha \geq 0$. Differentiate:

$$\frac{dL}{d\alpha} = 4 - 16\alpha = 0 \quad \Rightarrow \quad \alpha^\star = \frac{4}{16} = \frac{1}{4}.$$

Check $\alpha^\star \geq 0$ and it gives a concave quadratic maximum.

So the optimal dual variables are

$$\boxed{\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \tfrac{1}{4}}.$$

Recover primal w from stationarity:

$$w = \sum_{i=1}^{4} \alpha_i y_i x_i = \tfrac{1}{4}\big((+1)(1,1) + (+1)(1,-1) + (-1)(-1,1) + (-1)(-1,-1)\big).$$

Compute the vector inside parentheses:

$$(1,1) + (1,-1) + (1,-1) + (1,1) = (4,0),$$

hence

$$w = \tfrac{1}{4}(4,0) = (1,0).$$

Thus

$$\boxed{w = (1,0)^\top.}$$

Find b using KKT complementary slackness (support vectors satisfy equality $y_i(w^\top x_i + b) = 1$). Any training point can be used — take $x_1 = (1,1), y_1 = +1$:

$$1 \cdot (w^\top x_1 + b) = 1 \quad \Rightarrow \quad (1,0)\cdot(1,1) + b = 1 \quad \Rightarrow \quad 1 + b = 1,$$

so $b = 0$. Therefore

$$\boxed{b = 0.}$$

Finally verify the classifier: $f(x) = \mathrm{sign}(w^\top x + b) = \mathrm{sign}(x_1)$. Points with $x_1 = +1$ are classified $+1$, those with $x_1 = -1$ classified -1. The margin is $1/\|w\| = 1$.

Answer summary

$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \tfrac{1}{4}, \quad w = (1,0)^\top, \quad b = 0.$$

Question 4: Harder Kernel-Trick Question (Explicit cubic feature map)

Problem. Let $x = (x_1, x_2)^\top$ and $z = (z_1, z_2)^\top$ be vectors in $\mathbb{R}^2$. Consider the cubic polynomial kernel with bias

$$K(x,z) = (1 + x^\top z)^3 = \big(1 + x_1 z_1 + x_2 z_2\big)^3.$$

1. Expand $K(x,z)$ using the multinomial theorem and determine the dimensionality of the corresponding feature space.

2. Construct an explicit feature map $\phi : \mathbb{R}^2 \to \mathbb{R}^{10}$ (list the 10 components) such that

$$K(x,z) = \phi(x)^\top \phi(z).$$

   You should include the correct combinatorial (square-root) scaling factors so the dot product of $\phi(x)$ and $\phi(z)$ equals $K(x,z)$.

3. Verify the equality $K(x,z)=\phi(x)^\top\phi(z)$ by showing the dot product reproduces the expanded polynomial.

Answer

1. Multinomial expansion and dimension. Using the multinomial theorem with three terms $a=1$, $b=x_1z_1$, $c=x_2z_2$,

$$(1+x_1z_1+x_2z_2)^3=\sum_{i+j+k=3}\frac{3!}{i!j!k!}\,1^i(x_1z_1)^j(x_2z_2)^k,$$

where the sum runs over nonnegative integers i,j,k with $i+j+k=3$. The number of distinct monomials in the expansion equals the number of nonnegative integer solutions to $i+j+k=3$, which is $\binom{3+3-1}{3}=\binom{5}{3}=10$. Hence the feature space has dimension 10.

2. Explicit feature map $\phi(x)$. Group terms in the expansion by the monomials in $x=(x_1,x_2)$. A convenient orthonormalizing choice is to assign each monomial a component scaled by the square root of the corresponding multinomial coefficient so that $\phi(x)^\top\phi(z)$ reconstructs the coefficient in the expansion.

The 10 (ordered) multi-indices (i,j,k) with $i+j+k=3$ correspond to the following types of monomials in x:

degree 0 : $1\quad(i=3,j=0,k=0)$

degree 1 : $x_1,\ x_2\quad(i=2,j=1,k=0 \text{ and } i=2,j=0,k=1)$

degree 2 : $x_1^2,\ x_1x_2,\ x_2^2\quad(i=1,j=2,k=0;\ i=1,j=1,k=1;\ i=1,j=0,k=2)$

degree 3 : $x_1^3,\ x_1^2x_2,\ x_1x_2^2,\ x_2^3\quad(i=0,j=3,k=0;\ i=0,j=2,k=1;\ i=0,j=1,k=2;$ i=0,j=0,k=3).

For each term the multinomial coefficient is $\frac{3!}{i!j!k!}$. We choose components of $\phi(x)$ to be the corresponding monomial multiplied by the square root of this coefficient (so that the product of two such components gives the coefficient times $x^{\text{monomial}}z^{\text{monomial}}$).

Thus one valid ordering for $\phi(x)\in\mathbb{R}^{10}$ is:

$$\phi(x)=\begin{pmatrix}\sqrt{1}\cdot 1\\ \sqrt{3}\,x_1\\ \sqrt{3}\,x_2\\ \sqrt{3}\,x_1^2\\ \sqrt{6}\,x_1x_2\\ \sqrt{3}\,x_2^2\\ \sqrt{1}\,x_1^3\\ \sqrt{3}\,x_1^2x_2\\ \sqrt{3}\,x_1x_2^2\\ \sqrt{1}\,x_2^3\end{pmatrix}.$$

(Here the square roots come from $\sqrt{\frac{3!}{i!j!k!}}$; for example the coefficient of x_1x_2 arises from the case $(i,j,k)=(1,1,1)$ giving $\frac{3!}{1!1!1!}=6$, hence the component is $\sqrt{6}\,x_1x_2$.)

3. Verification. Compute $\phi(x)^\top\phi(z)$ by pairing corresponding components:

$$\begin{aligned}\phi(x)^\top\phi(z)=&\ (1)(1)\\ &+(\sqrt{3}x_1)(\sqrt{3}z_1)+(\sqrt{3}x_2)(\sqrt{3}z_2)\\ &+(\sqrt{3}x_1^2)(\sqrt{3}z_1^2)+(\sqrt{6}x_1x_2)(\sqrt{6}z_1z_2)+(\sqrt{3}x_2^2)(\sqrt{3}z_2^2)\\ &+(x_1^3)(z_1^3)+(\sqrt{3}x_1^2x_2)(\sqrt{3}z_1^2z_2)+(\sqrt{3}x_1x_2^2)(\sqrt{3}z_1z_2^2)+(x_2^3)(z_2^3).\end{aligned}$$

Collect coefficients; each product reproduces the corresponding multinomial coefficient times the monomial product. Rewriting this sum yields exactly the multinomial expansion of $(1+x_1z_1+x_2z_2)^3$. Hence

$$\phi(x)^\top\phi(z)=(1+x_1z_1+x_2z_2)^3=K(x,z),$$

as required.

Remarks.

- The general degree-p polynomial kernel with bias, $K(x,z) = (1+x^\top z)^p$, has feature map dimension $\binom{d+p}{p}$ for $x \in \mathbb{R}^d$. For $d=2, p=3$ this gives $\binom{5}{3} = 10$.
- The choice to use square-rooted multinomial coefficients in ϕ yields an explicit orthonormal-style mapping so that $\phi(x)^\top \phi(z)$ directly reproduces the multinomial coefficients in the kernel expansion.

Question 5: The Mechanics of Splitting (ID3)

Consider the following training dataset S for a binary classification problem where $Y \in \{0,1\}$. There are two binary features, $X_1 \in \{A, B\}$ and $X_2 \in \{C, D\}$.

Sample	X_1	X_2	Y
1	A	C	0
2	A	C	0
3	A	D	1
4	B	C	0
5	B	D	1
6	B	D	1

1. **Root Entropy:** Calculate the Entropy of the dataset $H(Y)$ before any split. Use base-2 logarithms.
2. **Information Gain:** Calculate the Information Gain $Gain(S, X_1)$ and $Gain(S, X_2)$. Which feature would the ID3 algorithm choose for the root node?
3. **Tree Construction:** Draw/Describe the full decision tree produced by the ID3 algorithm. Does the tree achieve 0 training error?
4. **Comparison:** The lecture slides mention that trees are built "Top-down". If we used a "Bottom-up" approach, would the resulting tree necessarily be identical? Explain briefly.

Answer

1. **Root Entropy Calculation (Baseline Uncertainty):** Before splitting the data, we must quantify the initial impurity (uncertainty) of the target variable Y across the entire dataset S.

   The dataset contains $N = 6$ samples. We observe the class distribution:

   - **Positive ($Y = 1$):** 3 samples. Probability $p_+ = 0.5$.
   - **Negative ($Y = 0$):** 3 samples. Probability $p_- = 0.5$.

   Since the classes are perfectly balanced, the entropy is at its mathematical maximum:

   $$\begin{aligned} H(S) &= -p_+ \log_2 p_+ - p_- \log_2 p_- \\ &= -0.5 \log_2(0.5) - 0.5 \log_2(0.5) \\ &= 1.0 \text{ bit} \end{aligned}$$

2. **Information Gain Calculation:** We evaluate each feature to determine which one reduces the uncertainty most effectively.

   Feature X_1 Analysis: Splitting on X_1 creates two subsets, but neither is pure.

   - $X_1 = A$: 3 samples with distribution $(1+, 2-)$.

   $$H(S_A) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} \approx 0.918 \text{ bits}$$

- $X_1 = B$: 3 samples with distribution $(2+, 1-)$.

$$H(S_B) \approx 0.918 \text{ bits}$$

The Information Gain is the expected reduction in entropy:

$$\begin{aligned} Gain(S, X_1) &= H(S) - \sum \frac{|S_v|}{|S|} H(S_v) \\ &= 1.0 - [0.5(0.918) + 0.5(0.918)] \\ &= \mathbf{0.082} \text{ bits} \end{aligned}$$

Feature X_2 Analysis: Splitting on X_2 creates a perfect separation of classes.

- $X_2 = C$: 3 samples, all Negative $(0+)$. Pure node $(H = 0)$.
- $X_2 = D$: 3 samples, all Positive $(3+)$. Pure node $(H = 0)$.

$$Gain(S, X_2) = 1.0 - 0 = \mathbf{1.0} \text{ bits}$$

Conclusion: The algorithm selects X_2 because $Gain(X_2) > Gain(X_1)$. Mathematically, X_2 provides a deterministic relationship with Y, eliminating all uncertainty.

3. **Tree Structure:** The tree makes the optimal split at the root using X_2.

   - **Root:** Test feature X_2.
     - If $X_2 = C$: Entropy is 0. We stop and output Leaf **0**.
     - If $X_2 = D$: Entropy is 0. We stop and output Leaf **1**.

   Since all leaf nodes have an entropy of 0, the model has perfectly memorized the training data, resulting in a Training Error of 0.

4. **Top-down vs. Bottom-up Approaches:** The resulting trees are not guaranteed to be identical due to the nature of the optimization.

   - **ID3 (Top-Down):** This is a **greedy** algorithm. It seeks a *local* optimum at every step (maximizing Gain for the current subset) without considering future splits. Once a split is made, it is never reconsidered.
   - **Bottom-Up:** A bottom-up approach would start by treating every data point as a leaf and merging them based on similarity or distance metrics (like hierarchical clustering).

   Mathematically, a sequence of locally optimal choices (Top-Down) does not always lead to the globally optimal structure that might be found by analyzing the data as a whole (Bottom-Up).

Question 6: Optimization Geometry & Jensen's Inequality

The Entropy function $H(p)$ is often visualized as a concave parabola. Let $H(p) = -p \log_2 p - (1-p)\log_2(1-p)$.

1. **Proof of Concavity:** Compute the second derivative $\frac{\partial^2 H(p)}{\partial p^2}$ and prove that Entropy is strictly concave for $p \in (0, 1)$.
2. **Jensen's Inequality:** Jensen's Inequality states that for a strictly concave function f: $\sum \lambda_i f(x_i) \leq f(\sum \lambda_i x_i)$. Use this to prove that Information Gain is always non-negative ($Gain \geq 0$).
3. **Optimization Trap:** The ID3 algorithm uses "Hill Climbing". Describe a scenario where this greedy approach gets stuck in a local optimum that is worse than the global optimum.

Answer

1. **Proof of Concavity:** To make the derivatives easier, let's first convert the base-2 logarithm to the natural logarithm using the identity $\log_2(x) = \frac{\ln x}{\ln 2}$. We can factor out the constant term $C = \frac{1}{\ln 2}$.

   The function becomes:

   $$H(p) = -C\left[p \ln p + (1-p)\ln(1-p)\right]$$

   Now, let's find the **First Derivative** using the product rule:

   $$\begin{aligned} H'(p) &= -C\left[(\ln p + \frac{p}{p}) + (\ln(1-p)\cdot(-1) + \frac{1-p}{1-p}\cdot(-1))\right] \\ &= -C\left[\ln p + 1 - \ln(1-p) - 1\right] \\ &= -C\ln\left(\frac{p}{1-p}\right) \end{aligned}$$

   Next, the **Second Derivative**:

   $$\begin{aligned} H''(p) &= -C\left[\frac{d}{dp}\ln p - \frac{d}{dp}\ln(1-p)\right] \\ &= -C\left[\frac{1}{p} - \frac{1}{1-p}\cdot(-1)\right] \\ &= -\frac{1}{\ln 2}\left[\frac{1}{p} + \frac{1}{1-p}\right] \\ &= -\frac{1}{\ln 2}\left[\frac{1}{p(1-p)}\right] \end{aligned}$$

   Conclusion: Because p represents a probability $(0 < p < 1)$, the term $p(1-p)$ is always positive. Since we have a negative sign in front, $H''(p)$ is always **negative**. In calculus, a negative second derivative everywhere proves the function is **Strictly Concave**.

2. **Proof of Non-Negative Gain:** Jensen's Inequality states that for any concave function (like Entropy), the "Function of the Average" is always greater than or equal to the "Average of the Functions."

   Let's map this to our decision tree:

   - Let p_{parent} be the overall probability of positives.
   - Let the parent split into children with weights λ_i and probabilities p_i.

   Mathematically, the parent's probability is just the weighted average of the children:

   $$p_{parent} = \sum \lambda_i p_i$$

   Applying Jensen's Inequality:

   $$\underbrace{H\left(\sum \lambda_i p_i\right)}_{\text{Entropy of Parent}} \geq \underbrace{\sum \lambda_i H(p_i)}_{\text{Weighted Entropy of Children}}$$

   Since Information Gain is defined as the entropy we *started with* minus the entropy we *ended up with*:

   $$Gain = H_{parent} - H_{weighted\_children}$$

   From the inequality above, the first term is larger than the second term. Therefore, the result must be non-negative ($Gain \geq 0$). This proves that splitting on a feature can never theoretically increase our uncertainty.

   item **Optimization Trap (The XOR Problem):** ID3 performs a greedy local search, selecting the attribute X_i that maximizes $Gain(S, X_i)$ at the current step. We can prove mathematically why this fails for the XOR function ($Y = X_1 \oplus X_2$).

   Proof: Consider the dataset for XOR: $\{(0,0,0), (0,1,1), (1,0,1), (1,1,0)\}$.

- **Step 1: Global Optimum Exists.** A tree of depth 2 that splits on X_1 and then X_2 perfectly classifies the data. The conditional entropy of this full tree is:

$$H(Y|X_1, X_2) = 0$$

  This represents the global minimum of the impurity function.

- **Step 2: Local Gradient is Zero.** To find this tree, ID3 must first select a root node. Let's calculate the Information Gain for X_1:

  Prior Entropy: $P(Y = 1) = 0.5 \implies H(S) = 1.0$.

  Conditional Entropy of X_1*:*
  - If $X_1 = 0$: $Y \in \{0, 1\}$ (1 of each). $H(Y|X_1 = 0) = 1.0$.
  - If $X_1 = 1$: $Y \in \{1, 0\}$ (1 of each). $H(Y|X_1 = 1) = 1.0$.

$$Gain(S, X_1) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v) = 1.0 - (0.5(1.0) + 0.5(1.0)) = 0$$

  By symmetry, $Gain(S, X_2) = 0$.

- **Conclusion:** The algorithm evaluates $\nabla Gain = 0$ for all features. Because the Mutual Information $I(Y; X_1) = 0$ and $I(Y; X_2) = 0$, the algorithm perceives these features as statistically independent of Y and stops building (or picks randomly). It fails to detect that the multivariate mutual information $I(Y; X_1, X_2) = 1$ bit. This mathematically demonstrates that the entropy landscape is non-convex, and the greedy descent gets stuck in a local optimum (the null tree).

Question 7: The Failure of Greedy Search (XOR)

Your lecture slides state that decision trees can express "any boolean function" but usually prefer simpler trees. Consider the classic XOR problem with 4 training examples:

X_1	X_2	Y
0	0	0
0	1	1
1	0	1
1	1	0

1. **The Greedy Trap:** Compute $Gain(S, X_1)$ and $Gain(S, X_2)$. Based on these values, explain why a standard ID3 algorithm might fail to start building the tree or pick a random feature.

2. **Existence:** Draw/Describe a decision tree of Depth 2 that perfectly classifies this data.

3. **Inductive Bias:** Mitchell defines the Inductive Bias of ID3 as a "Preference Bias" rather than a "Restriction Bias".

   (a) Define "Preference Bias".

   (b) Explain why the failure in part (1) illustrates that ID3 does not search the complete hypothesis space (like BFS-ID3), but specifically prefers trees with high Information Gain near the root.

Answer

1. **Calculation and Proof of Failure:** First, let's establish the entropy of the full dataset S. The target Y has two 0s and two 1s:

$$H(S) = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 1.0 \text{ bit}$$

   Now, we calculate the Information Gain for feature X_1. We partition S into $S_{X_1=0}$ and $S_{X_1=1}$.

- **Case $X_1 = 0$:** The labels are $\{0, 1\}$. $P(Y = 1|X_1 = 0) = 0.5$. $H(S_{X_1=0}) = 1.0$.
- **Case $X_1 = 1$:** The labels are $\{1, 0\}$. $P(Y = 1|X_1 = 1) = 0.5$. $H(S_{X_1=1}) = 1.0$.

Gain Calculation:

$$\begin{aligned} Gain(S, X_1) &= H(S) - \sum_{v \in \{0,1\}} \frac{|S_v|}{|S|} H(S_v) \\ &= 1.0 - [0.5(1.0) + 0.5(1.0)] \\ &= \mathbf{0} \end{aligned}$$

By symmetry, $Gain(S, X_2) = 0$.

Mathematical Analysis: The failure occurs because the variables X_1, X_2 and Y form an **XOR** relationship (Parity function). Mathematically, $P(Y = 1|X_1 = x) = P(Y = 1)$ for any value x. This means X_1 and Y are statistically independent *marginally*.

$$I(Y; X_1) = 0 \quad \text{and} \quad I(Y; X_2) = 0$$

However, they are not independent *jointly*:

$$I(Y; X_1, X_2) = H(Y) - H(Y|X_1, X_2) = 1.0 - 0 = 1.0$$

The greedy ID3 algorithm optimizes the marginal Mutual Information $I(Y; X_i)$ at each step. Since the gradient of the information gain is zero at the root, the algorithm perceives the features as noise and halts, failing to maximize the joint information.

2. **Existence of Solution (Depth 2 Tree):** Although ID3 cannot find it greedily, a tree exists that perfectly models $Y = X_1 \oplus X_2$:
   - **Root Node:** Split on X_1.
     - **Left Child ($X_1 = 0$):** Split on X_2.
       * If $X_2 = 0 \implies Y = 0$
       * If $X_2 = 1 \implies Y = 1$
     - **Right Child ($X_1 = 1$):** Split on X_2.
       * If $X_2 = 0 \implies Y = 1$
       * If $X_2 = 1 \implies Y = 0$

   This demonstrates that the function is representable within the hypothesis space.

3. **Inductive Bias Analysis:**
   (a) **Preference Bias vs. Restriction Bias:** To understand why ID3 failed, we must distinguish between two types of bias:
      - **Restriction Bias (Language Bias):** This occurs when the algorithm is strictly incapable of representing the correct solution. For example, a linear classifier *cannot* learn a circle. ID3 does **not** have this bias; the hypothesis space of Decision Trees is complete and includes the correct XOR tree.
      - **Preference Bias (Search Bias):** This occurs when the algorithm can represent the solution but chooses a different one based on a heuristic. ID3 has a **Preference Bias**: it searches the complete space but specifically favors trees that are short and have high information gain near the root.

   (b) **Why ID3 Fails (The Search Strategy):** The failure is due to the **Search Strategy**, not the model's capability.
      - **The Heuristic:** ID3 uses “Greedy Hill Climbing.” It only takes a step (adds a split) if that specific step reduces entropy *immediately*.
      - **The Trap:** In the XOR problem, the interaction between features is complex. Splitting on X_1 alone yields **Zero Information Gain**. Because ID3 does not look ahead to depth 2, it sees the gain of 0, assumes X_1 is noise, and terminates the search.
      - **Contrast:** A global search strategy (like Breadth-First Search) that checks all possible trees of depth 2 would find the optimal solution. ID3 fails because its greedy bias prevents it from traversing the "flat" part of the optimization landscape to reach the solution.